

The Inventory Packing Problem

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We study the problem of finding the minimum number of identical storage areas required to hold n items for which demand is known and constant. The replenishments of the items within a single storage area may be time phased so as to minimize the maximum total storage capacity required at any time. This is the *inventory-packing problem*, which can be considered as a variant of the well-known bin-packing problem, where one constraint is nonlinear. We study the worst-case performance of six heuristics used for that earlier problem since the recognition version of the inventory-packing problem is shown to be NP complete. In addition, we describe several new heuristics developed specifically for the inventory-packing problem, and also study their worst-case performance. Any heuristic which only opens a bin when an item will not fit in *any* (respectively, *the last*) open bin needs, asymptotically, no more than $25/12$ (resp., $9/4$) times the optimal number of bins. Improved performance bounds are obtainable if the range from which item sizes are taken is known to be restricted. Extensive computational testing indicates that the solutions delivered by these heuristics are, for most problems, very close to optimal in value.

The efficient storage of a commodity is a problem which is common to many industrial and commercial settings. For example, it may be appropriate in a production system to minimize the average quantity of raw materials, work in process, or finished goods. In designing a data base network, there may be considerations of access time and cost, as well as the cost of running each data base used. The modeling of such applications will depend upon how costs are incurred. For example, in many applications, the cost incurred is closely approximated by a linear function of the number of storage areas required. In the example of a production system, we may think of minimizing the number of units of warehouse space which are leased. In wishing to store a number of blocks of data with known size, it may be cost effective to minimize the number of identical data bases which are used. Such problems have attracted considerable attention in the operations research literature.

The most popular mathematical model for studying problems of the kind described in the *bin-packing problem*. This problem can be stated very simply: "given n items of sizes $R_1, \dots, R_n$, where $0 < R_i \leq 1, i = 1, \dots, n$, what is the minimum number of unit size bins required to hold those items?" The bin-packing problem is known to be NP hard, according to the classifications discussed in Garey and Johnson [10]. This classification has motivated the design and analysis of heuristic solution techniques for this problem. Indeed, the bin-packing problem is one of the most frequently studied problems in the field of operations research. Extensive surveys of results for this problem can be found in Garey and Johnson [11] and in Coffman et al. [7]. A very readable tutorial on proof techniques is that of Coffman [5]. Empirical studies (Johnson

[20], Hall et al. [16]) indicate that several heuristics routinely provide solutions which are very close to optimal for large instances of the bin-packing problem. In addition, there are a number of strong probabilistic results for the bin-packing problem (for example, Ong et al. [24], Rhee and Talagrand [27]) which suggest that under quite natural assumptions about the distribution of the item sizes, heuristic performance on large problems is very good.

The use of the bin-packing problem seems appropriate in the context of applications where the items to be packed remain fixed over time. However, where those items represent inventories of different products, it is likely that demand and periodic replenishment will frequently change the level of each product which needs to be stored. For concreteness, consider an n -item inventory problem in which demand is known and constant. If we consider an item individually, the inventory level of that item will be periodically replenished by production or purchase. It will then decrease linearly, as a result of demand, until the next replenishment. The bin-packing problem may be used to model the storage problem created by *simultaneous* replenishment, where R_i will represent the storage requirement of one replenishment of item i , $i = 1, \dots, n$. The assumption is made that it is important for efficient access and record keeping that the entire inventory of any item should be held within the same bin.

The focus of this article is on the advantages which can be gained by *not* replenishing all items simultaneously, but instead time-phasing those replenishments so that the maximum total storage requirement is reduced. As a small numerical example, consider a two-item problem in which a replenishment of each item requires 0.6 units of storage. Clearly, using simultaneous replenishment, the storage of those items will require two unit size storage areas, or bins. However, as will be demonstrated in section 1, the maximum total storage requirement of those items if their replenishments are suitably time-phased is 0.9 units. Consequently, this time-phasing permits both the items to be stored in a single bin of unit size.

The time-phasing of inventory replenishments, in order to utilize available storage space effectively, was introduced by Page and Paul [25]. The minimization of the maximum storage requirement in such a system is discussed by Zoller [32]. The assumption is made in both these articles that all the items have equal cycle times. Using a similar assumption, Hall [14] studied an n -item economic order quantity (EOQ) model (see, for example, Hadley and Whitin [12, Ch. 2]). The costs in that model included a cost determined by the maximum storage requirement. Inventory replenishment time-phasing problems of this type are called *inventory cycle balancing* problems. The concepts of inventory cycle balancing were used by Hall [15] to solve a variant of the well-known joint replenishment problem considered by Shu [29], Nocturne [23], and Jackson et al. [19]. Finally, a discrete-time inventory cycle balancing problem, where replenishment is only possible at k specific points in time, is shown to be NP hard (Hall [13]). However, a simple heuristic, based on the idea of rounding continuous-time solutions, provides both a good worst-case performance bound (which improves as k increases), and also consistently good performance over a large number of randomly generated problems.

The use of a restricted set of cycle times is very common in the literature of multi-item inventory theory, partly because it makes optimal closed-form solutions easier to find. For example, Hanssman [17] suggested the use of equal cycle times for multi-item problems. Elmaghraby [8] compared various approaches, and noted the ease with which a policy of equal cycle times provides feasible solutions. The use of a very

small and structured set of possible cycle times was justified by Maxwell and Singh [22]. In particular with regard to the present model, a policy of equal cycle times makes it easy to avoid very high peaks in total storage requirement, which may occur when items with unequal cycle times are replenished together. For example, if we use the popular alternative assumption that the cycle times for each item i is some integer k_i times the smallest cycle time (one unit), then it will happen occasionally that the replenishments of all the items will coincide. In fact, it follows from the Chinese remainder theorem (see Adams and Goldstein [1, p. 68]) that such an event will occur every $\text{lcm}(k_1, \dots, k_n)$ units of time. The storage requirement at that time is very large compared to others. Thus, if that requirement has to fall within some upper-bound constraint, it is likely that, for most of the planning horizon, storage requirements are well below constraints; i.e., available space is not being effectively utilized.

Returning to the issue of solvability, optimal closed-form solutions using the time-phasing procedure *without* the assumption of equal cycle times have so far only been obtainable for the special case of two items (Hartley and Thomas [18], Thomas and Hartley [31], Hall [15]). Moreover, we should not conclude that the equal cycle time assumption necessarily entails greatly increased cost. The computational results of Page and Paul [25] suggest that there are many situations in which the more effective utilization of space made possible by the time-phasing procedure outweighs, from a cost viewpoint, the restriction of the feasible set entailed by equal cycle times. Rosenblatt [28] proposed a simple method for discovering whether the equal cycle time method, or an alternative Lagrangian multiplier approach (Churchman et al. [4], Buchan and Koenigsberg [3], Hadley and Whitin [12], Starr [30]) works better for a multiple-item EOQ model with a single storage capacity constraint. Quite often the equal cycle time approach provides the optimal solution. A final advantage of equal cycle times is that they provide long-run repetitiveness of the manpower requirements related to the inventory cycle, which may in turn reduce the need for overtime.

The problem studied in this article has potential application beyond inventory modeling. A problem studied by Park and Yun [26] involves minimizing the maximum resource requirement needed in the performance of periodic activities. If the activities have common periodicity and linearly decreasing resource requirements, that problem is mathematically equivalent to the problem studied by Zoller [32]. The present model could be employed to find the minimum number of project teams, with known work rate, needed to fulfill a given set of tasks. Nevertheless, for consistency, we will continue to discuss the problem as a multi-item inventory problem.

The problem we address involves the minimization of the number of unit size bins, required by an infinite-horizon inventory-control policy which permits the time-phasing of replenishments in accordance with the idea of inventory cycle balancing. Demand is assumed to be known and constant for all items. We assume that backlogging is not permitted. The cycle time of all items is assumed to be equal, as discussed above. For items which are stored in separate bins, there is no advantage to time-phasing their replenishments. However, we will assume throughout that items stored within the same bin may have their replenishments time-phased. The specific details of this time-phasing will be discussed in section 1. Following that discussion, a mathematical formulation of the problem will be provided.

It is clear that the problem described contains features of both the bin-packing problem and of multi-item inventory-replenishment problems. For this reason, the problem is called the *inventory-packing problem*. This problem is very general, and

could be used to model any application satisfying the assumptions of the preceding paragraphs. However, the most likely applications are complex production-inventory systems for which demand is predictable and storage space is available in units of fixed size.

In section 1 of the article, we formalize the inventory-cycle balancing idea and provide a numerical example of its effectiveness in reducing the maximum total storage requirement of a multi-item inventory problem. Section 2 contains a description of the next-fit and first-fit heuristics for the inventory-packing problem, and provides upper and lower bounds on their worst-case performance ratios. A similar analysis for the next-fit decreasing and first-fit decreasing heuristics appears in section 3. Other heuristics, including several designed specifically for the inventory-packing problem, are considered in section 4. The effect on worst-case heuristic performance of restricting the item size is studied in section 5. All the heuristics described are subjected to extensive computational testing in section 6. That testing indicates that several of the heuristics routinely provide very close to optimal solutions to large, randomly generated instances of the inventory-packing problem. Finally, we provide some conclusions and make suggestions for further research.

1. INVENTORY CYCLE BALANCING

Assume there are n items ($n \geq 2$) which are replenished once per cycle. Let R_i represent the storage requirement of a replenishment of item i , $i = 1, \dots, n$. If all items are replenished simultaneously, the total storage requirement at that time will be $\sum_{i=1}^n R_i$. Assume, without loss of generality, that item 1 is replenished at time 0 within the cycle of length one unit of time. Let t_i denote the time of the next replenishment of item i , $i = 2, \dots, n$. The problem is to find the values of $t_2, \dots, t_n$ which minimize the maximum total storage requirement occurring over the cycle. The solution to this problem is now given.

LEMMA 1 (Page and Paul [25]): (a) Number the items $1, \dots, n$ arbitrarily. Then $t_1 = 0$ and

$$t_i = \sum_{k=2}^i R_k / \sum_{k=1}^n R_k, \quad i = 2, \dots, n. \quad (1)$$

(b) The minimum value of the maximum total storage requirement is

$$\sum_{j=1}^n \sum_{k=1}^j R_k R_j / \sum_{j=1}^n R_j. \quad (2)$$

Returning briefly to the example of the introductory section, the minimum value of the maximum total storage requirement of two items with requirements of 0.6 and 0.6 units is $[(0.6)^2 + (0.6)^2 + (0.6)^2] / (0.6 + 0.6) = 1.08 / 1.20 = 0.90 < 1$. Therefore those items can both be stored within a single bin, as illustrated in Figure 1.

From expression (2) and Figure 1, the increase in maximum total storage requirement when an item of storage requirement R_{n+1} is added is not generally R_{n+1} . The increase will depend upon the storage requirements of the other items sharing the same bin. From (2), that increase I is

$$I = \sum_{j=1}^{n+1} \sum_{k=1}^j R_k R_j / \sum_{j=1}^{n+1} R_j - \sum_{j=1}^n \sum_{k=1}^j R_k R_j / \sum_{j=1}^n R_j. \quad (3)$$

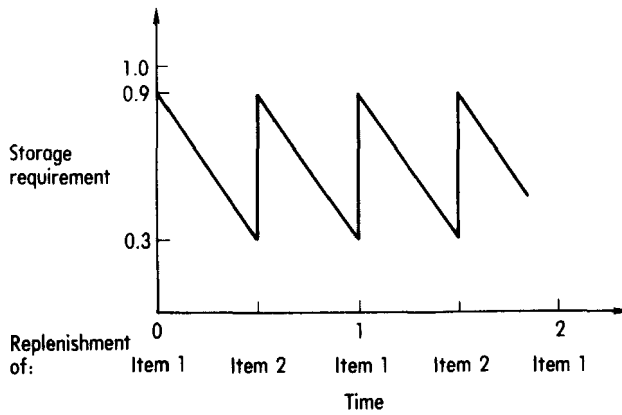


Figure 1. Inventory cycle balancing with two items.

When an item i is the first to be placed in a bin, $I = R_i$. Also, if the total storage capacity S of the bin satisfies

$$\sum_{j=1}^{n+1} \sum_{k=1}^j R_k R_j / \sum_{j=1}^{n+1} R_j > S,$$

then item $n + 1$ will not fit in that bin, and will need to be stored elsewhere. Throughout this article, except in discussing the complexity of the problem at Lemma 2, we assume that $S = 1$, i.e., the bins are of unit size, and also that $0 < R_i \leq 1$, $i = 1, \dots, n$. If several bins already contain at least one item, the choice of which of those bins to use for the next item may be made in several different ways. Thus the problem of minimizing the number of bins required in the inventory-packing problem seems to contain complications which go beyond those of the traditional bin-packing problem. Nevertheless, we may hope that bin-packing heuristics will still perform reasonably well for the inventory-packing problem.

A formulation of the inventory-packing problem is

$$[\text{IPP}] \quad \min \sum_{j=1}^n Z_j \quad (4)$$

$$\text{subject to} \quad \sum_{i=1}^n \sum_{k=1}^i R_i Y_{ij} R_k Y_{kj} / \sum_{i=1}^n R_i Y_{ij} \leq 1, \quad j = 1, \dots, n, \quad (5)$$

$$\sum_{i=1}^n Y_{ij} \leq M Z_j, \quad j = 1, \dots, n, \quad (6)$$

$$\sum_{j=1}^n Y_{ij} = 1, \quad i = 1, \dots, n, \quad (7)$$

$$Y_{ij} \in (0,1), \quad i, j = 1, \dots, n$$

$$Z_j \in (0,1), \quad j = 1, \dots, n.$$

In this formulation, R_i represents the size of item i , $i = 1, \dots, n$. Variable Y_{ij} has a value of one if item i is assigned to bin j . By previous assumption, no item size exceeds one, otherwise the problem would be infeasible. Variable Z_j takes on a value

of one if and only if bin j is used. M is a positive integer, ensuring that a bin must be opened before an item can be placed in it. We now need to define the complexity of problem [IPP].

LEMMA 2: The recognition version of problem [IPP] is NP complete.

PROOF: We rescale the data by clearing fractions, so that $R_1, \dots, R_n$ are integers, and the bin capacity (C) is increased in proportion. The recognition version of [IPP] is "given item storage requirements $R_1, \dots, R_n$ and an integer F , can those items be packed into no more than F bins, each having capacity C ?" It is easy to see that this problem is in NP. For the special case $F = 2$, this problem is equivalent to asking: "does there exist a partition S_1, S_2 of $S = \{1, \dots, n\}$ such that $\sum_{j \in S_k} \sum_{i \in S_k} R_i R_j / \sum_{j \in S_k} R_j \leq C, k = 1, 2$?" [IPPR]. We prove that the well-known NP-complete problem partition (Garey and Johnson [10, p. 223]) reduces to [IPPR]. Given an instance of partition defined by integer item sizes $R_1, \dots, R_n$, we wish to know whether there exists a partition S_1, S_2 of $S = \{1, \dots, n\}$ such that

$$\sum_{j \in S_k} R_j = \sum_{j=1}^n R_j / 2, \quad k = 1, 2.$$

The reduction is as follows. Define four new items numbered $n+1, n+2, n+3, n+4$ with integer size V , where

$$V = R_{n+1} = R_{n+2} = R_{n+3} = R_{n+4} \\ = \max \left\{ 3 \sum_{i=1}^n R_i, 2 \sum_{i=21}^n R_i^2 + \left(\sum_{i=1}^n R_i \right)^2 - \left\lfloor \sum_{i=1}^n R_i / 2 \right\rfloor + 1 \right\}$$

and let

$$C = \left[12V + \sum_{j=1}^n R_j + 2 \right] / 8.$$

We note that two of the new items $n+1, \dots, n+4$ will be in each of S_1 and S_2 in an optimal partition. Also, by algebraic manipulation we can show that

$$\sum_{j \in S_k} \sum_{\substack{i \in S_k \\ i \leq j}} R_i R_j / \sum_{j \in S_k} R_j = \sum_{j \in S_k} R_j / 2 + \sum_{j \in S_k} R_j^2 / 2 \sum_{j \in S_k} R_j. \quad (8)$$

We show that answering [IPPR] is equivalent to answering the recognition question for partition. Assume without loss of generality that $\sum_{j=1}^n R_j / 2$ is integer. Then, from (8), the answer to the partition question is "yes" if

$$\sum_{j \in S_k} \sum_{\substack{i \in S_k \\ i \leq j}} R_i R_j / \sum_{j \in S_k} R_j = \sum_{j=1}^n R_j / 4 + V + \sum_{j \in S_k} R_j^2 / 2 \sum_{j \in S_k} R_j, \quad k = 1, 2 \quad (9)$$

and “no” if

$$\sum_{j \in S_k} \sum_{\substack{i \in S_k \\ i \leq j}} R_i R_j \Big/ \sum_{j \in S_k} R_j \geq \sum_{j=1}^n R_j/4 + V + \sum_{j \in S_k} R_j^2/2 \sum_{j \in S_k} R_j + 1/2, \quad \text{for some } k \in (1,2). \quad (10)$$

We demonstrate that when S_k contains exactly two items of size V , we can approximate

$$\sum_{j \in S_k} R_j^2/2 \sum_{j \in S_k} R_j \text{ by } \left(V/2 - \sum_{j \in S_k} R_j/4 \right).$$

Let the items in S_k be those items numbered $1, \dots, t$, as well as two items of size V . It is easy to show algebraically that

$$\left(\sum_{i=1}^t R_i^2 + 2V^2 \right) \Big/ \left(\sum_{i=1}^t R_i + 2V \right) > V - \sum_{i=1}^t R_i/2.$$

By definition of V ,

$$\begin{aligned} V &> 2 \sum_{i=1}^n R_i^2 + \left(\sum_{i=1}^n R_i \right)^2 - \sum_{i=1}^n R_i/2 > 2 \sum_{i=1}^t R_i^2 + \left(\sum_{i=1}^t R_i \right)^2 - \sum_{i=1}^t R_i/2 \\ &\Rightarrow \sum_{i=1}^t R_i^2 + 2V^2 < V \sum_{i=1}^t R_i + 2V^2 - \left(\sum_{i=1}^t R_i \right)^2/2 - V \sum_{i=1}^t R_i + \sum_{i=1}^t R_i/4 \\ &+ V/2 \Rightarrow \left(\sum_{i=1}^t R_i^2 + 2V^2 \right) \Big/ \left(\sum_{i=1}^t R_i + 2V \right) < V - \sum_{i=1}^t R_i/2 + 1/4. \end{aligned}$$

Thus,

$$V/2 - \sum_{j \in S_k} R_j/4 < \sum_{j \in S_k} R_j^2/2 \sum_{j \in S_k} R_j < V/2 - \sum_{j \in S_k} R_j/4 + 1/8. \quad (11)$$

If (9) holds, then from the second inequality of (11),

$$\sum_{j \in S_k} \sum_{\substack{i \in S_k \\ i \leq j}} R_i R_j \Big/ \sum_{j \in S_k} R_j < \sum_{j=1}^n R_j/8 + 3V/2 + 1/8, \quad k = 1, 2$$

and we report a “yes” answer for [IPPR]. If (10) holds, then from the first inequality of (11),

$$\sum_{j \in S_k} \sum_{\substack{i \in S_k \\ i \leq j}} R_i R_j \Big/ \sum_{j \in S_k} R_j > \sum_{j=1}^n R_j/8 + 3V/2 + 1/2$$

for some $k \in \{1,2\}$, in which case we report a “no” answer for [IPPR]. Q.E.D.

2. NEXT-FIT AND FIRST-FIT HEURISTICS

We need some notation, in order to formalize the discussion of heuristic performance which follows. Let H denote a heuristic for the inventory-packing problem, H_I the number of bins used by that heuristic for input list I , and L_I^* the optimal number of bins into which list I can be packed. Let $W(H, I) = H_I/L_I^*$. Then the asymptotic worst case ratio of H , denoted W_H , is given by

$$W_H = \inf\{r \geq 1 \mid \text{for some } N > 0, W(H, I) \leq r \text{ for all } I \text{ with } L_I^* \geq N\}.$$

We also denote by $H_B(I)$ the number of bins used by heuristic H in the bin-packing problem for input list I , and by $L_B^*(I)$ the optimal number of bins used in that problem for list I .

The next-fit heuristic for the inventory-packing problem operates in a very similar way to the analogous heuristic for the traditional bin-packing problem. It requires slightly more computation, using (3), in the inventory-packing problem to see whether a bin has enough room for an item, than in the bin-packing problem. Nevertheless, this heuristic is still faster than other heuristics which we will describe, since only one bin is tested to see if an item will fit. We can describe the heuristic as follows.

Next-Fit Heuristic [NF]

- Step 0.* Order the items arbitrarily: set $i \leftarrow 1$.
- Step 1.* Place item i in the highest-numbered bin which has so far been opened. If that bin cannot accept item i , open a new bin.
- Step 2.* Set $i \leftarrow i + 1$. If $i \leq n$, go to Step 1. Otherwise, stop.

For the traditional bin packing problem, it is well known that $NF_B(I) \leq 2L_B^*(I) + 1$ (Coffman [5]). Example problems exist for which the heuristic performs arbitrarily close to this bound. We give a similar analysis for the next-fit heuristic for the inventory-packing problem.

THEOREM 1:

$$2.0 \leq W_{NF} \leq 2.25$$

PROOF: First, we prove the upper bound. Consider the following weighting scheme (Coffman [5]) for items. Let $W(R_j) = 1 + R_j$, $j = 1, \dots, n$. The purpose of this scheme is very simple. Informally, we claim that (a) each consecutive pair of bins packed by the next-fit heuristic (excluding the last bin, LB) contains a total weight of at least 2, and (b) no feasible packing can place items of total weight more than 9/4 in a single bin. Thus

$$NF_I \leq \sum_{j=1}^n W(R_j)R_j + 1 \quad (12)$$

and

$$L_I^* \geq 4 \sum_{j=1}^n W(R_j)R_j/9, \quad (13)$$

from which the upper bound on heuristic performance immediately follows. We begin by proving (12). Let the items in any *two consecutive* bins packed by next fit be numbered $1, \dots, p$. Clearly $\sum_{j=1}^p R_j > 1$, since the items were not packed into a single bin. Then, using (8),

$$\begin{aligned} \sum_{j=1}^p \sum_{k=1}^j R_k R_j / \sum_{j=1}^p R_j &= \sum_{j=1}^p R_j/2 + \sum_{j=1}^p R_j^2/2 \sum_{j=1}^p R_j > 1 \\ &\Rightarrow \sum_{j=1}^p R_j/2 + \sum_{j=1}^p R_j^2/2 > 1 \text{ since } \sum_{j=1}^p R_j > 1 \\ &\Rightarrow \sum_{j=1}^p R_j(1 + R_j) > 2 \\ &\Rightarrow \sum_{j=1}^p W(R_j)R_j > 2; \end{aligned}$$

therefore any two consecutive bins, excluding LB, have total weight greater than 2, and (12) follows.

To prove (13), observe that if a collection of items $1, \dots, q$ fits within a single bin, then

$$\sum_{j=1}^q \sum_{k=1}^j R_k R_j / \sum_{j=1}^q R_j \leq 1,$$

and therefore, using (8),

$$\sum_{j=1}^q R_j + \sum_{j=1}^q R_j^2 / \sum_{j=1}^q R_j \leq 2. \quad (14)$$

Now consider the problem:

$$\text{maximize } f(q) = \sum_{j=1}^q R_j + \sum_{j=1}^q R_j^2,$$

subject to (14) and $0 < R_j \leq 1, j = 1, \dots, q$.

We now prove $f(q) \leq 9/4$. The upper bounds on R_j can be ignored, since even a single $R_j > 1$ violates constraint (14). We use the classical Lagrangian (λ) method to obtain first order conditions

$$\begin{aligned} 1 + 2R_k - \lambda - 2\lambda R_k / \sum_{j=1}^q R_j + \lambda \sum_{j=1}^q R_j^2 / \sum_{j=1}^q R_j &= 0, \\ \Rightarrow \lambda &= (1 + 2R_k) \sum_{j=1}^q R_j / \left(\sum_{j=1}^q R_j + 2R_k - \sum_{j=1}^q R_j^2 \right), k = 1, \dots, n, \end{aligned} \quad (15)$$

and, since $\lambda \neq 0$,

$$\sum_{j=1}^q R_j + \sum_{j=1}^q R_j^2 / \sum_{j=1}^q R_j = 2. \quad (16)$$

Solving any pair of equations (15) with $k = i$ and $k = t$, we conclude that

$$(R_i - R_t) \left(\sum_{j=1}^q R_j^2 - \sum_{j=1}^q R_j + 1 \right) = 0, \quad 1 \leq i < t \leq q. \quad (17)$$

Consider in turn the two alternative solutions available from (17). Solving

$$\sum_{j=1}^q R_j^2 - \sum_{j=1}^q R_j + 1 = 0$$

simultaneously with (16), we find that

$$\left(\sum_{j=1}^q R_j \right)^2 - \sum_{j=1}^q R_j - 1 = 0$$

and then

$$\sum_{j=1}^q R_j = (1 + \sqrt{5})/2, \quad \sum_{j=1}^q R_j^2 = (\sqrt{5} - 1)/2, \text{ and } f(q) = \sqrt{5}.$$

Alternatively, from (17), $R_1 = \dots = R_q$. From expression (2), $2/(q + 1)$ is the largest size at which q identical items will fit in the same bin, therefore we set $R_1 = \dots = R_q = 2/(q + 1)$. Then $f(q) = 2q/(q + 1) + 4q/(q + 1)^2 = (2q^2 + 6q)/(q + 1)^2$, and thus $f(q) - f(q - 1) = (2q^2 + 6q)/(q + 1)^2 - [2(q - 1)^2 + 6(q - 1)]/q^2 = (4 + 6q - 2q^2)/q^2(q + 1)^2$. It follows that $f(q)$ is a unimodal concave function of the integer values of q , which achieves its maximum at $q = 3$. Also, $f(3) = 6/4 + 12/16 = 9/4 > \sqrt{5}$. Thus if items $1, \dots, q$ are packed within the same bin, $\sum_{j=1}^q W(R_j)R_j \leq 9/4$, proving (13). The following example proves the lower bound.

EXAMPLE 2.1: Consider the input list L of $4n$ items, $L = 2/3, \delta, 2/3, \delta, 2/3, \delta, \dots$, where $\delta > 0$ is small. The next-fit heuristic packs each consecutive pair of items into a single bin, thus $NF_L = 2n$. But the optimal packing consists of n bins, each containing two items of size $2/3$, and one bin containing all the other items, giving $L_L^* = n + 1$. Therefore $W_{NF} \geq \lim_{n \rightarrow \infty} [2n/(n + 1)] = 2$. Q.E.D.

Next, we study the first-fit heuristic which considers not only the most recently opened bin but also earlier bins, when trying to pack an item.

First Fit Heuristic [FF]

- Step 0.* Order the items arbitrarily, set $i \leftarrow 1$.
- Step 1.* Place item i in the lowest numbered bin in which it fits. If it does not fit in any open bin, open a new bin.
- Step 2.* Set $i \leftarrow i + 1$. If $i \leq n$, repeat Step 1. Otherwise, stop.

For the traditional bin-packing problem, Johnson et al. [21] show that $FF_B(I) \leq 1.7L_B^*(I) + 2$. An example problem shows that the asymptotic worse-case ratio of 1.7 can be attained. We now give a similar analysis for the first-fit heuristic for the inventory-packing problem.

THEOREM 2:

$$1.71 \leq W_{FF} \leq 2.083.$$

PROOF: To analyze the first-fit heuristic, we use a slightly different weighting function. Let $V(R_j) = 1 + 3R_j/4$, $j = 1, \dots, n$. We claim

$$FF_f \leq \sum_{j=1}^n V(R_j)R_j + 1 \quad (18)$$

and

$$L_f^* \geq 12 \sum_{j=1}^n V(R_j)R_j/25 \quad (19)$$

proving the upper bound. It is easy to show, using (2), that if the total size of items in a bin is no more than $2/3$, then a further item of size $2/3$ or less can also fit in the bin. Therefore, if any bin contains items of total size not exceeding $2/3$, each higher-numbered bin must contain at least one item of size $2/3$ or more. Thus, there is at most one bin (LB) containing items of total size less than $2/3$. Let the items in any two consecutive bins, excluding LB, be numbered $1, \dots, p$. Clearly, $\sum_{j=1}^p R_j \geq 4/3$. Then, using (8),

$$\begin{aligned} \sum_{j=1}^p \sum_{k=1}^j R_k R_j / \sum_{j=1}^p R_j &= \sum_{j=1}^p R_j/2 + \sum_{j=1}^p R_j^2/2 \sum_{j=1}^p R_j > 1 \\ \Rightarrow \sum_{j=1}^p R_j/2 + 3 \sum_{j=1}^p R_j^2/8 &> 1 \text{ since } \sum_{j=1}^p R_j \geq 4/3 \\ \Rightarrow \sum_{j=1}^p R_j(1 + 3R_j/4) &> 2 \\ \Rightarrow \sum_{j=1}^p V(R_j)R_j &> 2; \end{aligned}$$

therefore any two consecutive bins, excluding LB, have total weight greater than 2, and (18) follows.

Now consider the problem:

$$\text{maximize } g(q) = \sum_{j=1}^q R_j + 3 \sum_{j=1}^q R_j^2/4$$

$$\text{subject to (14) and } 0 < R_j \leq 1, \quad j = 1, \dots, q.$$

We now prove $g(q) \leq 25/12$. The first order conditions generate (17) as in the proof of Theorem 1. Thus

$$\sum_{j=1}^n R_j = (1 + \sqrt{5})/2, \quad \sum_{j=1}^n R_j^2 = (\sqrt{5} - 1)/2,$$

and $g(q) = (1 + \sqrt{5})/2 + 3(\sqrt{5} - 1)/8 = 2.08156$.

Alternatively, from (17), $R_1 = \dots = R_q = 2/(q + 1)$ and $g(q) = 2q/(q + 1) + 3q/(q + 1)^2 = (2q^2 + 5q)/(q + 1)^2$. Then $g(q) - g(q - 1) = (2q^2 + 5q)/(q + 1)^2 - [2(q - 1)^2 + 5(q - 1)]/q^2 = (3 + 5q - q^2)/(q + 1)^2$. The maximum

value is $g(5) = 5/3 + (3/4)(5/9) = 25/12 > 2.08156$. Thus if items $1, \dots, q$ are packed within the same bin,

$$\sum_{j=1}^q V(R_j)R_j \leq 25/12,$$

proving (19). The following example proves the lower bound.

EXAMPLE 2.2: Let $L = 1/51 + \delta, \dots, 2/7 + \delta, \dots, 1/2 + \delta, \dots, 2/3 + \delta, \dots$, where $\delta > 0$ is small, and there are $100n$ items of each size. Then $FF_I = 100n/100 + 100n/5 + 100n/2 + 100n = 171n$. But one item of each size will fit in a bin, therefore $L_I^* \leq 100n$, and $W_{FF} \geq \lim_{n \rightarrow \infty} (171n/100n) = 1.71$. Q.E.D.

It is interesting to note that the asymptotic worst-case ratio of the first-fit heuristic for the inventory-packing problem is not equal to 1.7 as in the bin-packing problem.

As Coffman et al. [7] indicate, better bin-packing heuristic performance, measured by worst-case performance guarantees or by indicators of average performance, can be achieved by ordering the list of items by nonincreasing size. The theoretical implications of this procedure for the inventory-packing problem are examined in the next section.

3. NEXT-FIT DECREASING AND FIRST FIT DECREASING HEURISTICS

The next-fit decreasing (NFD) heuristic is the same as next fit, except that Step 0 is replaced by:

Step 0'. Order the items by nonincreasing size; set $i \leftarrow 1$.

The worst-case performance of NFD is now analyzed.

THEOREM 3:

$$1.71 \leq W_{NFD} \leq 2.08\bar{3}.$$

PROOF: First we prove the upper bound. Observe that the items packed into a bin are packed in order of nonincreasing size. If the first item has size of $2/3$ or more, the bin will immediately have a total item size of at least $2/3$. If the size of the first item packed is less than $2/3$, then so are all subsequent item sizes; therefore further items can be added until the total item size is at least $2/3$. Only the last bin (LB) can have total item size less than $2/3$. Let the items in any two consecutive bins, excluding LB, be numbered $1, \dots, p$. Clearly, $\sum_{j=1}^p R_j \geq 4/3$. Let $V(R_j) = 1 + 3R_j/4$, $j = 1, \dots, n$. The proof now follows that of Theorem 2. The following example proves the lower bound.

EXAMPLE 3.1: Let $L = 2/3 + \delta, \dots, 1/2 + \delta, \dots, 2/7 + \delta, \dots, 1/51 + \delta, \dots$, where there are $100n$ items of each size and where $\delta > 0$ is small. Then

$NFD_I = 100n + 100n/2 + 100n/5 + 100n/100 = 171n$. However, one item of each size will fit in a bin, therefore $L_I^* \leq 100n$ and $W_{NFD} \geq \lim_{n \rightarrow \infty} (171n/100n) = 1.71$. Q.E.D.

Since the asymptotic worst-case ratio of the next-fit decreasing heuristic for bin packing is 1.691 . . . (Baker and Coffman [2]), the same ratio does not apply here. The first-fit decreasing [FFD] heuristic follows the first-fit procedure, but with Step 0 replaced by Step 0' above. It is also possible to analyze the worst-case performance of this heuristic for the inventory-packing problem.

THEOREM 4:

$$1.25 \leq W_{FFD} \leq 2.08\bar{3}.$$

PROOF: The upper bound follows from Theorem 2. It is not easy to find example problems for which the first-fit decreasing heuristic performs badly. Indeed, as the results in section 6 indicate, that heuristic routinely provides solutions which are very close to optimal. The worst performance so far discovered is illustrated by the following example problem.

EXAMPLE 3.2: Let $L = 2/3 + \delta, \dots, 2/5 + \epsilon, \dots, 2/5 + \gamma, \dots$, where there are $6n$ items of size $2/3 + \delta$, $6n$ items of size $2/5 + \epsilon$, and $12n$ items of size $2/5 + \gamma$, and where $\delta, \epsilon, \gamma > 0$, $\epsilon > \gamma$, are chosen as follows. We require that three items of sizes $2/3 + \delta$, $2/5 + \gamma$, and $2/5 + \gamma$ will fit in a single bin, but that three items of sizes $2/3 + \delta$, $2/5 + \epsilon$, and $2/5 + \gamma$ will not fit in a bin. Then the first-fit decreasing heuristic packs $6n$ bins with one item of size $2/3 + \delta$ and one item of size $2/5 + \epsilon$, and $4n$ bins with three items of size $2/5 + \gamma$. Hence $FFD_I = 6n + 4n = 10n$. But one possible packing is $6n$ bins, each containing one item of size $2/3 + \delta$ and two items of size $2/5 + \gamma$, and $2n$ bins each containing three items of size $2/5 + \epsilon$. Thus $L_I^* \leq 6n + 2n = 8n$ and $W_{FFD} \geq \lim_{n \rightarrow \infty} (10n/8n) = 1.25$. Q.E.D.

The asymptotic worst-case ratio here is more than for the first-fit decreasing heuristic when applied to the bin-packing problem. The result in that case is a bound of 1.222 (Johnson et al. [21]).

4. OTHER HEURISTICS

As observed by Fisher [9] in general, and in particular for the bin-packing problem by Johnson [20] and Hall [13], worst-case performance guarantees may not be a good indicator of typical heuristic performance. Consequently, it is necessary to perform computational testing in order to assess the typical performance. A natural place from which to begin this testing is with the next-fit and first-fit heuristics discussed earlier, and with the corresponding versions for lists ordered by decreasing item size.

Among other heuristics which seem worth testing are best fit, and its corresponding version with items ordered by nonincreasing storage requirement size, best fit decreasing. It is particularly important that this last heuristic should be tested because the computational results of Johnson [20] suggest that best fit decreasing and first fit

decreasing provide the equal best average performance, for randomly generated problems, over a variety of heuristics. The best-fit heuristic is now described.

Best Fit Heuristic [BF]

- Step 0.* Order the items arbitrarily; set $i \leftarrow 1$.
- Step 1.* Place item i in the fullest open bin in which it will fit. If it will not fit in any open bin, open a new bin.
- Step 2.* Set $i \leftarrow i + 1$. If $i \leq n$, repeat Step 1. Otherwise, stop.

For the best-fit decreasing [BFD] heuristic, Step 0 is replaced by Step 0' above. We now have available six heuristics from the traditional bin-packing problem. It is also possible to design heuristics specifically for the inventory-packing problem. In particular, we again note that when an item i is added to a bin in the inventory-packing problem, the increase in the storage requirement of that area is usually less than R_i . Furthermore, that increase depends upon the other items already in that bin. We make use of this fact to design two highly intuitive heuristics for the inventory-packing problem. The first of these is called the minimum increase heuristic.

Minimum Increase Heuristic [MI]

- Step 0.* Order the items arbitrarily; set $i \leftarrow 1$.
- Step 1.* Place item i in the bin which (among all the open bins in which it will fit) has the minimum increase I in storage requirement as a result of item i being added. In the event of a tie, place item i in the lowest-numbered open bin in which it will fit. If it will not fit in any open bin, open a new bin.
- Step 2.* Set $i \leftarrow i + 1$. If $i \leq n$, repeat Step 1. Otherwise, stop.

This heuristic has no analog for the traditional bin-packing problem, because in that problem the effect on the total storage requirement resulting from adding an item is not dependent upon items already held in the same bin. This comment would also apply to the next heuristic described.

Minimum Unused Proportion Heuristic [MUP]

- Step 0.* Order the items arbitrarily; set $i \leftarrow 1$.
- Step 1.* Place item i in the bin which (among all the open bins in which it will fit) has the minimum value for the ratio of the resulting increase I in storage requirement to the amount of unused storage requirement prior to adding item i . In the event of a tie, place item i in the lowest-numbered bin in which it will fit. If it will not fit in any open bin, open a new bin.
- Step 2.* Set $i \leftarrow i + 1$. If $i \leq n$, repeat Step 1. Otherwise, stop.

The intuition behind the minimum increase heuristic is the "greedy" notion of paying the smallest "price," in this case measured by increase in storage requirement, to perform a task. As the number of items in a bin increases, the increase in storage requirement which would result from adding a given item will tend to fall; thus the minimum increase heuristic will tend to fill up any open bins very quickly. However, it is likely that the minimum unused proportion heuristic will favor open bins with fewer items, since the amount of unused storage requirement will be greater in such bins. Consequently, the two heuristics are likely to produce different solutions. Because

previous computational studies of the bin-packing problem (Johnson [20], Hall et al. [16]) have concluded that a list of items ordered by decreasing size is often easier to pack, we also tested versions of the minimum increase and minimum unused proportion heuristics which incorporate such an ordering. As with the best-fit heuristic, Step 0 is in both cases replaced by Step 0'. These two heuristics are named, respectively, the minimum increase decreasing [MID] and minimum unused proportion decreasing [MUPD] heuristics. The worst-case performance of the heuristics described in this section is now studied.

THEOREM 5:

- (a) $1.71 \leq W_{BF} \leq 2.08\bar{3}$,
- (b) $1.25 \leq W_{BFD} \leq 2.08\bar{3}$,
- (c) $1.71 \leq W_{MI} \leq 2.08\bar{3}$,
- (d) $1.25 \leq W_{MID} \leq 2.08\bar{3}$,
- (e) $1.71 \leq W_{MUP} \leq 2.08\bar{3}$,
- (f) $1.25 \leq W_{MUPD} \leq 2.08\bar{3}$.

PROOF: The upper bound for (a) to (f) follows from Theorem 2. The lower bound for (a), (c), and (e) follows from Example 2.2. The lower bound for (b), (d), and (f) follows from Example 3.2. Q.E.D.

Since Johnson et al. [21] proved an asymptotic worst-case bound of $1.22\bar{2}$ for the best-fit decreasing heuristic, we observe at Theorem 5(b) that the ratio of that heuristic is worse for the inventory packing problem.

5. RESTRICTION OF ITEM SIZES

For the bin-packing problem, there are numerous results (see, for example, Coffman et al. [5]) for worst-case performance when the items are all generated from the interval $(0, \Theta]$, where $\Theta < 1$. Intuitively, such results are interesting because it is often the larger items which are difficult to pack efficiently, and which create examples of poor heuristic performance. A similar analysis for the inventory-packing problem yields results for all the heuristics which have been described. These results are now summarized.

THEOREM 6: For any heuristic H for the inventory-packing problem, let $W_H(\Theta)$ denote the asymptotic worst-case ratio (as defined in section 2), when all items are generated from the interval $(0, \Theta]$, where $\Theta < 1$. Let G denote a generic heuristic which assigns items sequentially to bins without repacking, which only opens a new bin when an item will not fit in any previously opened bin, and which does not order the items before packing. Thus G could be any of four heuristics, i.e., FF, BF, MI, and MUP. Let GD denote a heuristic which works similarly, except that items may be ordered by decreasing size prior to packing, i.e., FFD, BFD, MID, and MUPD. Let $L(\Theta)$ denote the set of all lists of items with every item size in the interval $(0, \Theta]$ where $\Theta \leq 1/2$, and let $k = \lfloor 2/\Theta \rfloor - 1$. Then for all lists from $L(\Theta)$

- (a) $W_{NF}(\Theta), W_G(\Theta), W_{NFD}(\Theta), W_{GD}(\Theta) \leq 2(k + 1)/(2k - \Theta k - \Theta)$
- (b) $W_{NF}(\Theta) \geq k/(k - 1)$

Table 1. Performance bounds for item sizes in the interval $(0, \theta]$

θ	k	Upper bound	$W_{NF}(\theta) \geq$	$W_G(\theta) \geq$	$W_{NFD}(\theta) \geq$	$W_{GD}(\theta) \geq$
0.500	3	2.000	1.500	1.250	1.250	1.166
0.450	3	1.905	1.500	1.250	1.250	1.166
0.400	4	1.667	1.333	1.200	1.200	1.150
0.300	5	1.464	1.250	1.166	1.166	1.133
0.250	7	1.333	1.166	1.125	1.125	1.107
0.200	9	1.250	1.125	1.100	1.100	1.088
0.100	19	1.112	1.055	1.050	1.050	1.047

- (c) $W_G(\Theta) \geq (k + 2)/(k + 1)$
- (d) $W_{NFD}(\Theta) \geq (k + 2)/(k + 1)$
- (e) $W_{GD}(\Theta) \geq (k^2 + 2k - 1)/(k^2 + k)$.

PROOF: (a) From expression (2), it is possible to pack k items each of size $2/(k + 1)$ into a single bin, giving a total item size of $2k/(k + 1)$ in that bin. Therefore any collection of items, each of size $\leq \theta \leq 2/(k + 1)$, which does not fit in a single bin, must have total item size $> 2k/(k + 1)$. The largest possible contribution of any one item to the total item size is Θ . Consequently, any bin which is not used again must contain items of total size $\geq 2k/(k + 1) - \Theta$. Therefore the generic heuristic can have at most one bin with total item size $< 2k/(k + 1) - \Theta = (2k - k\Theta - \Theta)/(k + 1)$. Since no bin can contain items of total size exceeding 2, the result follows. The proof of (b) through (e) is by a series of example problems.

(b) $L = 2/(k + 1), \dots, \delta, 2/(k + 1), \dots, \delta$, where $\delta > 0$ is small, and where there are n sequences each consisting of $(k - 1)$ items of size $2/(k + 1)$ followed by one item of size δ . If δ is small, then $NF_I = n(k - 1)/(k - 1) = n$ and $L_I^* \leq n(k - 1)/k + 1$; therefore $W_{NF}(\Theta) \geq k/(k - 1)$.

(c) $L = 2/(k + 3) + \delta, \dots, 2/(k + 2) + \delta, \dots$, where $\delta > 0$ is small, and where there are n items of the smaller size and kn items of the larger size. Then $G_I = n/(k + 1) + n = n(k + 2)/(k + 1)$. But k items of the larger size and one of the smaller size will fit in a single bin, therefore $L_I^* \leq n$. Thus $W_G(\Theta) \geq (k + 2)/(k + 1)$.

(d) $L = 2/(k + 2) + \delta, \dots, 2/(k + 3) + \delta, \dots$, where $\delta > 0$ is small, and where there are kn items of the larger size and n items of the smaller size. Then $NFD_I = kn/k + n/(k + 1) = n(k + 2)/(k + 1)$, but $L_I^* \leq n$ as at (c). Thus $W_{NFD}(\theta) \geq (k + 2)/(k + 1)$.

(e) $L = 2/(k + 2) + \delta, \dots, 2/(k + 2) - \beta, \dots$, where there are $n(k - 1)$ items of the larger size, and $2n$ items of the smaller size, and where $\delta, \beta > 0$ are chosen as follows. We require that $(k - 1)$ items of the larger size and 2 items of the smaller size will fit in a single bin, but that k items of the larger size and 1 item of the smaller size will not fit in a single bin. Then $GD_I = n(k - 1)/k + 2n/(k + 1) = n(k^2 + 2k - 1)/k(k + 1)$ and $L_I^* \leq n$, therefore $W_{GD}(\Theta) \geq (k^2 + 2k - 1)/(k^2 + k)$. Q.E.D.

Some sample values of the upper and lower bounds on the asymptotic worst-case performance ratio resulting from Theorem 6 are shown in Table 1.

In Table 1, each row represents a different value for the maximum item size, Θ .

Table 2. Mean heuristic error ratios.

n	TOPSIZ	NF	NFD	FF	FFD	BF	BFD	MI	MID	MUP	MUPD
80	1.0	1.305	1.313	1.033	1.047	1.073	1.153	1.033	1.031	1.034	1.033
80	0.9	1.307	1.353	1.050	1.037	1.077	1.160	1.033	1.033	1.033	1.033
80	0.75	1.213	1.310	1.033	1.033	1.067	1.117	1.033	1.033	1.033	1.033
120	1.0	1.314	1.314	1.038	1.030	1.064	1.138	1.019	1.020	1.018	1.015
120	0.9	1.298	1.354	1.034	1.026	1.068	1.148	1.020	1.022	1.022	1.020
120	0.75	1.228	1.314	1.020	1.020	1.052	1.106	1.021	1.020	1.020	1.020
160	1.0	1.305	1.311	1.029	1.023	1.056	1.149	1.012	1.013	1.012	1.015
160	0.9	1.295	1.359	1.026	1.024	1.055	1.141	1.013	1.013	1.013	1.019
160	0.75	1.224	1.308	1.019	1.015	1.043	1.103	1.014	1.013	1.013	1.021
200	1.0	1.305	1.311	1.023	1.017	1.044	1.135	1.011	1.013	1.013	1.011
200	0.9	1.291	1.356	1.020	1.013	1.053	1.133	1.013	1.013	1.013	1.015
200	0.75	1.224	1.307	1.020	1.013	1.040	1.113	1.014	1.013	1.013	1.013
Mean Error ratio		1.276	1.326	1.029	1.025	1.058	1.133	1.020	1.020	1.020	1.020

In each row, $k = \lfloor 2/\Theta \rfloor - 1$ represents the maximum number of items of size exactly Θ which can be placed in a single bin. The next column shows the common upper bound on $W_{NF}(\Theta)$, $W_G(\Theta)$, $W_{NFD}(\Theta)$, and $W_{GD}(\Theta)$. The last four columns show lower bounds on, respectively, $W_{NF}(\Theta)$, $W_G(\Theta)$, $W_{NFD}(\Theta)$, and $W_{GD}(\Theta)$. In a number of cases, the upper-bound ratios represent improvements (for special data sets) over general results presented earlier. The lower-bound results are shown in order to provide a measure of accuracy for the upper-bound results.

6. COMPUTATIONAL RESULTS

In order to assess the typical performance of the ten heuristics which have been described, a large number of random test problems were generated. Heuristic results were validated by comparing the values delivered for each problem with the optimal value, which was found by the use of a specially designed enumerative (branch-and-bound) algorithm. Whereas the computation time required by the optimal algorithm increased rapidly with problem size and approached 90 minutes of Prime 5500 time for the largest problems, the operation of the heuristics required less than 5 seconds in all cases. This relative advantage in the speed of heuristic computation increases with problem size.

A total of 240 random problems were generated, consisting of 12 series of 20 problems each. The testing of the ten heuristics for each problem resulted in a total of 2400 observations. The parameters of each problem were the number of items (n) and the maximum storage requirement (TOPSIZ). The data for a problem thus consists of n items, each with a storage requirement generated from the uniform distribution over the interval $(0, \text{TOPSIZ}]$.

The heuristic error ratio, which is the number of bins required by the heuristic divided by the optimal number of bins required, was found in all cases. Table 2 contains details of the mean heuristic ratio, for each of the ten heuristics, calculated over the 20 randomly generated problems within each series. Each row of Table 2 represents a series of problems, and each column represents a heuristic. The NF, NFD, BF, and BFD heuristics show less effective typical performance than the other six

Table 3. Maximum mean heuristic error ratios.

n	TOPSIZ	NF	NFD	FF	FFD	BF	BFD	MI	MID	MUP	MUPD
80	1.0	1.367	1.400	1.067	1.033	1.100	1.200	1.033	1.033	1.033	1.033
80	0.9	1.400	1.433	1.067	1.067	1.100	1.200	1.033	1.033	1.033	1.033
80	0.75	1.267	1.367	1.033	1.033	1.100	1.200	1.033	1.033	1.033	1.033
120	1.0	1.380	1.360	1.040	1.040	1.080	1.160	1.020	1.020	1.020	1.040
120	0.9	1.340	1.400	1.040	1.040	1.080	1.080	1.020	1.020	1.020	1.020
120	0.75	1.280	1.360	1.020	1.020	1.080	1.120	1.020	1.020	1.020	1.020
160	1.0	1.350	1.350	1.038	1.038	1.088	1.175	1.013	1.013	1.013	1.025
160	0.9	1.325	1.388	1.038	1.025	1.063	1.163	1.013	1.013	1.013	1.025
160	0.75	1.275	1.338	1.025	1.025	1.063	1.113	1.013	1.013	1.013	1.025
200	1.0	1.340	1.327	1.027	1.020	1.053	1.147	1.013	1.013	1.013	1.013
200	0.9	1.307	1.367	1.027	1.020	1.060	1.147	1.013	1.013	1.013	1.013
200	0.75	1.260	1.347	1.027	1.020	1.053	1.140	1.020	1.013	1.013	1.013
Mean Error ratio		<u>1.400</u>	<u>1.433</u>	<u>1.067</u>	<u>1.067</u>	<u>1.100</u>	<u>1.200</u>	<u>1.033</u>	<u>1.033</u>	<u>1.033</u>	<u>1.040</u>

heuristics tested. The relatively poor performance of BFD is something of a surprise, in view of the successful results obtained by Johnson [20] for the bin-packing problem. As expected, FF and particularly FFD perform very well for problem [IPP], just as they seem to for the traditional bin-packing problem. However, even better performance was achieved by the four heuristics designed specially for [IPP]. Over a large number of random problems, the average error was only about 2% for those heuristics, indicating that they are extremely effective in providing near-optimal solutions to large instances of [IPP]. It seems fair to point out that there is slightly more computation involved in operating the specially designed heuristics, because of the need to compute the increase in total storage requirement using (3), than there is in using the other heuristics. The range over which item sizes are generated seems to make no discernible difference to heuristic accuracy. However, it is clear from Table 2 that average heuristic performance improves as problem size increases.

The maximum error ratio observed over the 20 problems in each series is shown in Table 3. Once again, the NF, NFD, BF, and BFD heuristics seem less reliable than the other six heuristics tested. For the MI, MID, MUP, and MUPD heuristics, the worst result obtained over 240 randomly generated problems gave an error of less than 3.4%. Since the computation of the MI and MID heuristics is slightly faster and simpler than that of MUP and MUPD, the single heuristic which is most recommended (based on average and worst observed performance) by the computational study is MI. The minimum increase heuristic offers additional flexibility, in that it does not require the items to be ordered before processing. Thus it can be used in an environment where there are considerations external to the model which suggest that certain orderings of replenishments of items are better than others. Examples of such external considerations could be bottlenecks in material handling or in access to storage areas. Alternatively, it could be used in a situation where assignments of some items to bins must be made before all the item sizes are known. A similar variant of the bin-packing problem has been studied by Coffman et al. [6].

7. CONCLUSIONS AND POSSIBLE EXTENSIONS

The inventory-packing problem discussed in this article models the efficient storage of items in a multi-item inventory environment, where demand is known and constant.

The recognition version of this problem is shown to NP complete, suggesting that the exact solution of large problem instances will be very difficult. However, several heuristic approaches, borrowed from the closely related bin-packing problem, provide feasible solutions which use a number of bins guaranteed to be no more than a fixed constant times the optimum number of bins. That constant improves as the maximum item size falls below $1/2$. Moreover, some of those heuristics, and some additional heuristics designed specially for the inventory-packing problem, routinely provide very close to optimal solutions for large, randomly generated problem instances. Computational results suggest that typical heuristic performance improves as problem size increases.

Future work includes finding worst-case performance bounds which are tight (i.e., attainable). An important, although algorithmically difficult, generalization of the inventory-packing problem would involve the optimization of cycle times for the different items. This may be achieved under a variety of assumptions (e.g., integer multiples, powers-of-two) about the relative cycle times of different items, and using a variety of traditional inventory models (e.g., EOQ, joint replenishment). A different generalization involves packing items into bins of a number of distinct sizes, where the use of a larger bin incurs a larger cost. That model would seem slightly more realistic for many production-inventory systems. Another possible extension involves allowing some splitting of the storage of an item, in order to make better use of available space, at the possible expense of additional material handling and record keeping costs.

Since it contains elements of traditional inventory models, and of the well-studied bin-packing problem, interest in the inventory-packing problem goes beyond its potential applications. Particularly encouraging for further study is the successful performance, measured both by theoretical performance guarantees and by average behavior, of several bin-packing heuristics.

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